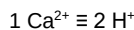


2022 JC2 Preliminary Examination
H2 Chemistry 9729
Paper 1 Worked Solution

1 $n_{\text{H}^+} = n_{\text{OH}^-} = \frac{25}{1000} \times 1.0 \times 10^{-2} = 2.50 \times 10^{-4} \text{ mol}$



$n_{\text{Ca}^{2+}} = \frac{1}{2} n_{\text{H}^+} = 1.25 \times 10^{-4} \text{ mol}$

$[\text{CaSO}_4] = [\text{Ca}^{2+}] = \frac{n_{\text{Ca}^{2+}}}{V} = \frac{1.25 \times 10^{-4}}{\frac{50}{1000}} = 2.50 \times 10^{-3} \text{ mol dm}^{-3}$

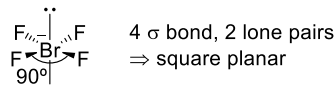
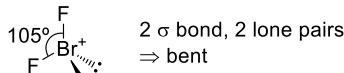
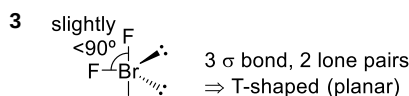
⇒ A

2 Large jump between 7th and 8th I.E.

⇒ 7 electrons in valence shell

⇒ Group 17

⇒ C



⇒ C

4 $pV = nRT$

$pV = \frac{m}{M} RT$

Plot of pV against T will be a **straight line** that passes through the **origin (0 K)** with

gradient $\frac{mR}{M}$.

Since $M(\text{E}) < M(\text{F})$, the gradient for **E** will be **steeper** than the gradient for **F**.

⇒ A

5 $\Delta G = \Delta H - T\Delta S$

1 *: $\Delta H > 0$, $\Delta S < 0$, $\Delta G > 0$ (non-spontaneous) at all T

2 *: $\Delta H < 0$, $\Delta S < 0$, $\Delta G > 0$ (non-spontaneous) when $T > \frac{\Delta H}{\Delta S}$

3 ✓: $\Delta H < 0$, $\Delta S > 0$, $\Delta G < 0$ (spontaneous) at all T

⇒ D

6 A ✓: 1 mol of H_2O is formed from the reaction of an acid and a base

B *: $\Delta H_{\text{lattice energy}}^{\text{HCO}_3^-}(\text{NaHCO}_3(\text{s}))$ is for the reaction $\text{Na}^+(\text{g}) + \text{HCO}_3^-(\text{g}) \rightarrow \text{NaHCO}_3(\text{s})$

C ✓: Being a strong acid, the HSO_4^- ion will fully dissociate in water to give $\text{H}^+(\text{aq})$ and $\text{SO}_4^{2-}(\text{aq})$

D ✓: $\Delta H_{\text{formation}}^{\text{H}_2\text{SO}_4}$ involves forming $(\text{NH}_4)_2\text{SO}_4$ from the constituent elements, S, H_2 , N_2 and O_2 in their standard states

⇒ B

7 1 ✓: From the slow step,
 $\text{rate} = k[(\text{CH}_3)_2\text{C}^+(\text{OH})][\text{H}_2\text{O}]$
 $= k'[(\text{CH}_3)_2\text{C}(\text{=O})][\text{H}_3\text{O}^+]$, $k' = kK_c$
 At low pH, $[\text{H}_3\text{O}^+]$ is high and is essentially a constant.
 $\text{rate} = k''[(\text{CH}_3)_2\text{C}(\text{=O})]$

2 *: rate constant, $k = Ae^{-\frac{E_a}{RT}}$ depends only on the E_a and T .

3 ✓: I_2 is only involved in a fast step after the slow (rate determining) step. Hence rate of reaction is independent of $[\text{I}_2]$.

⇒ D

8 A *: Since there is no change in the number of gaseous particles upon reaction, the total pressure in the vessel remains constant. Hence the amount of reactants and products cannot be determined.

B *: Slow cooling allows the equilibrium to readjust to the new temperature. Hence, the K_c determined will be that at 20 °C and not 500 °C.

C ✓: Rapid cooling will slow down the reaction, allowing the amount of I_2 in the equilibrium mixture at 500 °C, and hence K_c , to be determined.

D *: If the HI present is decomposed back to H_2 and I_2 , the amount of reactants and product in the equilibrium mixture cannot be determined.

⇒ C

9 A *: Since $\Delta H < 0$, position of equilibrium lies more to the left with increasing T .

Hence $K_p = \frac{1}{p_{\text{G(g)}}} \downarrow$ es with $\uparrow$ ing T .

B *: $pV = nRT \Rightarrow p = \frac{n}{V} RT = CRT$

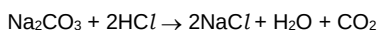
$K_p = \frac{1}{p_{\text{G(g)}}} = \frac{1}{[\text{G(g)}]RT} = \frac{K_c}{RT}$

C *: θ depends only on k_1 , k_2 and p_{G} . It is independent of the total surface area of the catalyst.

D ✓: At high p_{G} , $1 + \frac{k_1}{k_2} p_{\text{G}} \approx \frac{k_1}{k_2} p_{\text{G}}$. Hence,
 $\theta \approx 1$, which is independent of T .

⇒ D

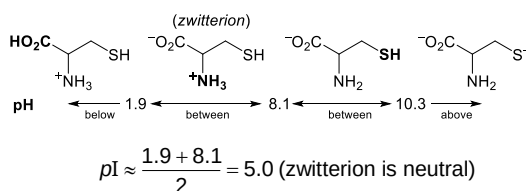
10 As HCl is titrated against Na_2CO_3 , Na_2CO_3 is added from the burette into HCl in the conical flask, the pH increases with the volume of titrant. HCl will *completely* react with any Na_2CO_3 added



Hence there will be only one end-point.

⇒ A

11 Considering the species present:



⇒ B

12 A *: $[\text{Ag}^+] = 2\sqrt[3]{\frac{K_{\text{sp}}(\text{Ag}_2\text{CO}_3)}{4}} = 2.53 \times 10^{-4}$

B *: $[\text{Ag}^+] = \sqrt{K_{\text{sp}}(\text{AgCl})} = 1.34 \times 10^{-5}$

C ✓: $[\text{Ag}^+] = \sqrt{K_{\text{sp}}(\text{AgBr})} = 7.35 \times 10^{-7}$

D *: $[\text{Ag}^+] = \sqrt{K_{\text{sp}}(\text{AgIO}_3)} = 1.79 \times 10^{-4}$

⇒ C

13 Residue: SiO_2 (insoluble in H_2O & H_2SO_4)

Filtrate: acidic (gives H_2 with Zn) $\Rightarrow \text{HCl}$

⇒ B

14 Reducing power (chemical reactivity) of Group 2 metals $\uparrow$ es down the group, since $\Rightarrow$ atomic radius $\uparrow$ es

$\Rightarrow \downarrow$ attraction of nucleus for valence e^- s

$\Rightarrow$ easier to loss valence e^- s

Atomic radius increases as the number of electron shells $\uparrow$ es, rendering valence electrons further from nucleus.

⇒ D

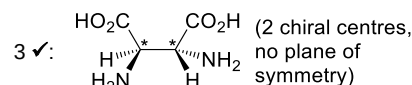
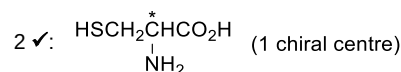
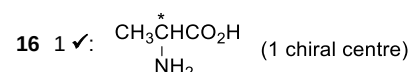
15 A *: Electronegativity $\downarrow$ es with $\uparrow$ ing size down Group 17, due to weaker attraction for shared pair of electrons.

B *: Oxidising power (tendency to get reduced) $\downarrow$ es with $\uparrow$ ing size down Group 17, due to weaker attraction for incoming e^- .

C ✓: Polarisability of X^- $\uparrow$ es with size down Group 17 as electrons are generally further from the nucleus.

D *: Oxidising power $\downarrow$ es (B) down Group 17, hence $E^\ominus(\text{X}_2|\text{X}^-)$ becomes less positive ($\downarrow$ es).

⇒ C



⇒ D

17 A *: Homolytic fission also occurs in the second step where the $\text{X}-\text{O}^\bullet$ bond in $\text{X}-\text{O}^\bullet$ is broken to give $\text{X}^\bullet$.

B *: Termination steps involves the removal of $\text{X}^\bullet$ through reaction with another radical.

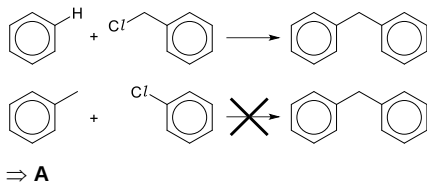
C ✓: $\text{X}^\bullet$ serves as a catalyst to speed up the breakdown of O_3 into O_2 .

D *: $\text{CCl}_2\text{F}_2 \rightarrow \bullet\text{CClF}_2 + \bullet\text{Cl}$ instead as the $\text{C}-\text{F}$ bond is much stronger than the $\text{C}-\text{Cl}$ bond.

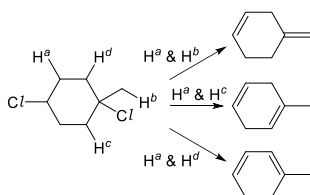
⇒ C

- 18 1 ✓: C=C in 2-methylbuta-1,3-diene reacts with Br₂(aq)
 2 ✗: There are no chiral elements (centres) in 2-methylbuta-1,3-diene
 3 ✗: The C=C-C=C π network is highly electron-rich and will repel approaching nucleophiles.
 ⇒ A

- 19 The two benzene rings can only be joined together via a *Friedel-Crafts alkylation* involving the aromatic hydrogen and a halogenoalkane:

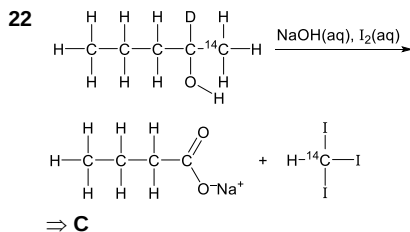


- 20 Both Cl undergoes elimination with excess ethanolic KOH:



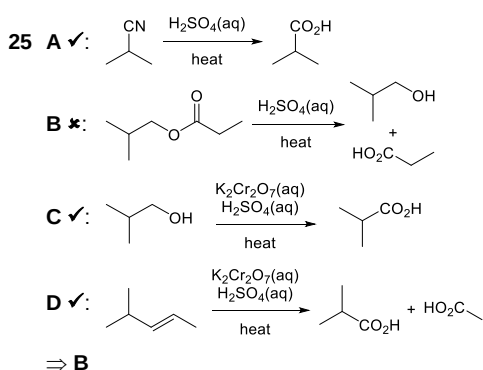
- 21 Halogenoarenes do not undergo nucleophilic substitution with ethanolic AgNO₃.

- A: 1 mol of AgCl (143.4 g)
 B: 1 mol of AgI (234.8 g)
 C: 1 mol of AgCl + 1 mol of AgI (378.2 g)
 D: 2 mol of AgBr (375.6 g)
 ⇒ C



- 23 R gives a red ppt with Fehling's solution
 ⇒ R is an aldehyde
 Oxidation of Q give R, an aldehyde
 ⇒ Q is a 1° alcohol
 ⇒ B

- 24 The carboxylic acid group reacts with alcohols (but not phenols) to give esters.
 CH₃CHO and C₆H₅COC₂H₅ both reacts with the amino group.
 ⇒ C



- 26 A ✗: The form of asparagine in aqueous solution depends on the pH of the solution. It exists solely in the form of the zwitterion at the isoelectric point.
 B ✓: The side chain, -CH₂CONH₂, being an *amide* is neutral.
 C ✗: Hot dilute H₂SO₄ hydrolyses the amide side chain to give NH₄⁺ (aq) and not NH₃(g)
 D ✗: All the 26 amino acids are crystalline solids at room temperature due to strong ionic bonding between the zwitterions.
 ⇒ B

27 [O]: Fe³⁺ + e⁻ ⇌ Fe²⁺ $E^{\ominus} = +0.77 \text{ V}$
 [R]: O₂ + 4H⁺ + 4e⁻ ⇌ 2H₂O $E^{\ominus} = +1.23 \text{ V}$

$$E_{\text{cell}}^{\ominus} = E_{\text{reduction}}^{\ominus} - E_{\text{oxidation}}^{\ominus}$$

$$= E^{\ominus}(\text{O}_2|\text{H}_2\text{O}) - E^{\ominus}(\text{Fe}^{3+}|\text{Fe}^{2+})$$

$$= +1.23 - (+0.77) = +0.46 \text{ V}$$

$$E_{\text{cell}} = E_{\text{reduction}} - E_{\text{oxidation}} = E(\text{T}) - E(\text{S})$$

- 1 ✗: [Fe²⁺] and [Fe³⁺] ↓es by the same extent. Eqm is not affected.
 $E(\text{S}) = E^{\ominus}(\text{Fe}^{3+}|\text{Fe}^{2+})$. No change to E_{cell} .
 2 ✓: CN⁻ forms complex with Fe³⁺ and Fe²⁺. New eqm established
 $[\text{Fe}(\text{CN})_6]^{3-} + e^- \rightleftharpoons [\text{Fe}(\text{CN})_6]^{4-}$
 with $E^{\ominus} = +0.36 \text{ V}$.
 $E(\text{S}) < E^{\ominus}(\text{Fe}^{3+}|\text{Fe}^{2+})$. Hence E_{cell} ↑es (more positive)
 3 ✗: [H⁺] ↓es. Eqm shifts left.
 $E(\text{T}) < E^{\ominus}(\text{O}_2|\text{H}_2\text{O})$ Hence E_{cell} ↓es (less positive)
 ⇒ D

- 28 Cathode: 2H₂O + 2e⁻ → H₂ + 2OH⁻
 Anode: 2H₂O → O₂ + 4H⁺ + 4e⁻
 $Q = It = 15.0 \times (5 \times 60) = 4500 \text{ C}$

$$n_{e^-} = \frac{Q}{F} = \frac{4500}{96500} = 0.04663 \text{ mol}$$

$$n_{\text{gas}} = n_{\text{O}_2} + n_{\text{H}_2} = \frac{1}{4}n_{e^-} + \frac{1}{2}n_{e^-} = \frac{3}{4}n_{e^-}$$

$$= \frac{3}{4} \times 0.04663 = 0.03497 \text{ mol}$$

⇒ C

- 29 density = $\frac{\text{mass}}{\text{volume}} \approx \frac{\text{mass of an atom}}{\text{volume of an atom}} \propto \frac{A_r}{r^3}$
- 1 ✓: A_r ↑es across the Period 4 TM, while *r* remain relatively constant, hence density ↑es
 2 ✓: Number of nucleons (protons and neutrons) ↑es ⇒ mass of atom ↑es, hence density ↑es
 3 ✗: ↑ in number of outer shell electrons has negligible effect on mass of the atom ($m_e \approx \frac{1}{1833}m_p$ or m_n)
 ⇒ B

- 30 Within the tetradentate ligand, the two N atoms with 3 bonds (and 1 lone pair) are neutral, but the two nitrogen with only 2 bonds (and 2 lone pairs) are negatively charged. Hence the tetradentate ligand has a -2 charge.

The remaining two ligands, CH₃OH and , are neutral. Hence for an

overall charge of +1 for the complex cation, Co must be in the +3 oxidation state.
 ⇒ C

Answer Key

Qn	Ans	Qn	Ans	Qn	Ans
1	A	11	B	21	C
2	C	12	C	22	C
3	C	13	B	23	B
4	A	14	D	24	C
5	D	15	C	25	B
6	B	16	D	26	B
7	D	17	C	27	D
8	C	18	A	28	C
9	D	19	A	29	B
10	A	20	B	30	C